

- BC 78. The radius of a circle is decreasing at a constant rate of 0.1 centimeter per second. In terms of the circumference C , what is the rate of change of the area of the circle, in square centimeters per second?

1998

6C

- (A) $-(0.2)\pi C$
(B) $-(0.1)C$
(C) $-\frac{(0.1)C}{2\pi}$
(D) $(0.1)^2 C$
(E) $(0.1)^2 \pi C$

- BC 20. When $x = 8$, the rate at which $\sqrt[3]{x}$ is increasing is $\frac{1}{k}$ times the rate at which x is increasing. What is the value of k ?

1998

NC

- (A) 3 (B) 4 (C) 6 (D) 8 (E) 12

- BC 5. If $3x^2 + 2xy + y^2 = 2$, then the value of $\frac{dy}{dx}$ at $x = 1$ is

1969

NC

- (A) -2 (B) 0 (C) 2 (D) 4 (E) not defined

5. Consider the curve given by $y^2 = 2 + xy$.

2005 B

- BC NC (a) Show that $\frac{dy}{dx} = \frac{y}{2y - x}$.

(b) Find all points (x, y) on the curve where the line tangent to the curve has slope $\frac{1}{2}$.

(c) Show that there are no points (x, y) on the curve where the line tangent to the curve is horizontal.

(d) Let x and y be functions of time t that are related by the equation $y^2 = 2 + xy$. At time $t = 5$, the value of y is 3 and $\frac{dy}{dt} = 6$. Find the value of $\frac{dx}{dt}$ at time $t = 5$.